THE UNITED REPUBLIC OF TANZANIA

ORGANISATION OF HEADS OF NON-GOVERNMENT SECONDARY SCHOOLS (OHONGSS)

KIGOMA REGIONS

FORM FOUR MOCK EXAMINATION

(ACTUAL PRACTICAL)

032/2A **CHEMISTRY 2A**

Time: 2:30 HOURS **YEAR: 2024**

Instructions

1. This paper consists of two (2) questions.
2. Answer all questions.
3. Each question carries 25 marks, making total of 50 marks.
4. Cellular phones and any other unauthorized materials are not allowed in the examination room.
5. Write your examination number on every page of your answer sheet.
6. You may use the following constants;
   H=1, Na= 23, O=16, S=32, Cl=35.5, C=12
   1litre=1dm^3=1000cm^3

MsomiBora.com

1. You are provided with the following solutions:
   AA: containing 5.2g of impure sulphuric acid in $1dm^3$ of solution
   BB: containing 4g of pure sodium hydroxide in $1dm^3$ of solution
   MO: methyl orange indicator
   (a) Pipette $25cm^3$ or $20cm^3$ of solution BB and titrate with the acid from the burette using two drops of the indicator and obtain three titre values
   (b) (i)…….cm^3 of BB required……cm^3 of AA for complete reaction.
   ii. The color changed from …….to …….....
   iii. Write a balanced chemical equation for the reaction at the neutral point.
   (c) Is the use of methyl orange indicator in this experiment as suitable as the use of phenolphthalein indicator? Give a reason for your answer.
   (d) (i). Explain how you will rinse the apparatus (burette and pipette) before doing the titration.
   (ii). Can litmus paper be used as an indicator in this experiment? Justify your answer.
   (e) Showing your procedures clearly calculate the percentage purity of sulphuric acid.

2. You are provided with the following solutions:
   M: solution containing 20g of sodium thiosulphate crystals dissolved in 1 litre of solution.
   N: 2M Hydrochloric acid solution
   D: Distilled water
   A stop watch, a piece of plain paper with a mark X, a 100ml beaker and measuring cylinder.
   Procedure:
   i. Place 100ml beaker on top of mark X on the piece of plain paper. Using a measuring cylinder, measure $30cm^3$ of solution M and pour it into a beaker.
   ii. Measure $10cm^3$ of solution N using different measuring cylinder and pour it into the beaker containing solution M and simultaneously start the stop watch. Watch the mixture from the above and observe the changes. Immediately stop the stop watch when the mark x is obscured and record the time taken for the mark X to be obscured.
   iii. Repeat the whole procedure (i) and (ii) using $25cm^3$, $20cm^3$, $15cm^3$ and $10cm^3$ of the original M and making the total volume up to $30cm^3$ by adding D. Record the results as shown in the table below.

Volume of M (cm^3)	Volume N (cm^3)	Volume of distilled water(cm^3)	Time taken for the mark X to be obscured(seconds)
30	10	0	
25	10	5	
20	10	10	
15	10	15	
10	10	20	

a. Copy and complete filling table above
b. Write down a balanced chemical equation for the reaction between M and N.
c. Write a net ionic equation for the molecular reaction in (b) above.
d. What make mark x to be obscured?
e. Using the data in the table, plot a graph of volume of M (cm^3) against time (sec.)
f. Write two uses of the product which obscured the mark X.
g. Suggest the aim of this experiment.